Can FOMC Sentences Labeled With Macroeconomic Policy Stances Better Predict Market Direction Than Traditional Sentiment Labels Alone?

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***Abstract*—The Federal Open Market Committee (FOMC) is the governmental body that sets monetary policy in the United States (US). Changes to the federal funds rate and the money supply can have significant impacts on the US economy. Thus, FOMC communications from meeting minutes, speeches, and press conferences can be valuable sources of textual data that contain insights into future monetary policy adjustments through stance signals. This research project aims to build on existing applications of NLP techniques in FOMC communications and to evaluate the usefulness of FOMC policy signal in predicting stock market movements. Multiple BERT-family transformer models and LLMs are used to classify FOMC sentences from meeting minutes based on their perceived hawkish/dovish monetary policy stance signal. The classifier with the best metrics is utilized to generate stance labels in a separate corpus for use in a 3-day S&P 500 market direction prediction experiment. The Claude Sonnet 4.6 model exhibits the best macro F1 and precision scores in the stance classification task. The 3-day market direction prediction results present no clear improvement when including an LLM-generated stance feature. The paper features robustness tests that evaluate market prediction results in separate time periods and prediction windows and the effect of input truncation on stance classification accuracy. The project's most significant finding is that its results are constrained by the lack of human-annotated FOMC sentence datasets. LLM-based data augmentation is proposed as a potential solution to create synthetic data for enhanced model training and class balancing.**

***Index Terms*—Federal Open Market Committee (FOMC), Large Language Models (LLM), FinBERT, FinBERT-FOMC, transformers, Monetary Policy, Dovish, Hawkish**

I. INTRODUCTION

A. Problem Statement

The Federal Open Market Committee (FOMC) is a division of the United States (US) Federal Reserve System (Fed) that dictates monetary policy through the management of the federal funds rate and the US money supply [1]. The changes to interest rates made by the FOMC can have significant impacts on equity and fixed-income markets, as well as the macroeconomy [2]. Significant research interest exists in analyzing FOMC communications, as they contain opinions and commentary on economic conditions from the body that sets monetary policy in the US. Prior research has shown that extracting and utilizing opinions and stances from FOMC textual data can improve macroeconomic predictions and financial decision-making [1], [3].

Performing sentiment analysis on FOMC communications is challenging due to the domain-specific language used and difficult sentence structures with conflicting sentiments [3]. Additionally, researchers and/or experts typically annotate FOMC text with traditional sentiments labels (positive/negative) that may misinterpret the Fed's underlying policy posture [3]. These challenges, along with the limited use of transformer-based models, present an opportunity to explore enhancements in the sentiment analysis of FOMC communications using domain-specific labels, fine-tuned transformer models, and LLMs to generate improved financial and macroeconomic predictions.

B. Research Questions

This research project aims to answer the following questions:

- How do domain-specific, fine-tuned transformer models perform against rule-based approaches and generic LLMs in classifying the policy stance of FOMC sentences using domain-specific labels (hawkish/dovish)?
- Is there an observed improvement in short-term (3-day) stock market directional movement prediction when including a domain-specific stance feature as opposed to a traditional sentiment feature using binary classification?

C. Literature Review

Existing literature concerning the sentiment analysis of financial news and FOMC communications largely consists of rule-based approaches and artificial neural networks (ANNs). [4] performs sentiment analysis on International Monetary Fund communications using a lexicon-based approach while [5] applies a similar lexicon-based approach on Fed communications. In contrast, [6], [7], and [8] use ANNs to perform sentiment analysis on financial news, market data, and social media. [9] implements a transformer-based BERT model to conduct sentiment analysis on Chinese financial news to better predict the CSI 300 index. [10] builds on [9] by utilizing a FinBERT model, a transformer-based model pre-trained on financial text, to classify the sentiment of global financial news. [1] and [11] utilize fine-tuned, domain-specific transformer models (FinBERT and FinBERT-FOMC) to perform sentiment analysis exclusively on Fed communications. [2], [3], and [12] go beyond previously mentioned works by performing sentiment analysis on Fed communications using LLMs and by conducting a comparative analysis of LLM performance against ANNs and fine-tuned transformer models. [3] classifies FOMC communications using domain-specific stance labels (hawkish/dovish) to obtain insight into the Fed's policy stance. In contrast, [2] provides traditional sentiment labels (positive/negative) for three pertinent criteria: growth, employment, and inflation. [6], [7], and [9] all observe improved stock market prediction accuracy when including sentiment from financial texts as feature(s) in their predictive models.

D. Gap Discussion

Prior researchers have successfully employed transformer-based sentiment analysis on financial and FOMC textual data to generate improved classification results over traditional methods. Although researchers have used transformer models, these works typically use traditional sentiment labels (positive/negative) that can fail to accurately capture policy insight (hawkish/dovish) from the Fed. Additionally, comparisons of traditional and domain-specific sentiment measures in financial market and macroeconomic predictive models remain limited. This research project expands upon prior works by comparing the stance classification performance of rule-based, fine-tuned transformer, and LLM models with domain-specific (hawkish/dovish) labels. Furthermore, this study evaluates the efficacy of including a domain-specific stance measure in the prediction of short-term (3-day) stock market directional movements.

II. Methodology

A. Data Acquisition and Analysis

Four datasets were used as part of the research project, with two datasets containing preprocessed, labeled FOMC sentences from [2] and [3]. The dataset (manual-mm-split.xlsx) produced by [3] containing sentences from FOMC meeting minutes was used to answer RQ1. The dataset consisted of the sentence and its annotated hawkish/dovish stance label. The researchers preprocessed the dataset by filtering out unrelated sentences based on a dictionary of key terms, sampled labeling, and sentence splitting. The dataset was used for the training and evaluation of the selected models in the RQ1 stance classification task.

The baseline dataset used to answer RQ2 was created by the researchers in [2]. The FOMC (FOMC.xlsx) dataset created by [2] consisted of medium length sentences from FOMC meeting minutes ranging from 2006 to 2024. The project utilized the preprocessed dataset created by the researchers, which contained annotated growth, employment, and inflation sentiment labels. The dataset was used in the LLM labeling pipeline, as the the LLMs were prompted to add hawkish/dovish stance labels as an additional feature to answer RQ2. S&P 500 price data (SPY ETF), inflation data (CPI), and unemployment (UNRATE) data were merged with the FOMC.xlsx dataset based on the closest upcoming FOMC meeting event date. Furthermore, 3 and 10-day market results and a market direction indicator were merged into the FOMC.xlsx dataset.

Additional preprocessing steps were performed on both datasets. Duplicate sentences, sentences with invalid labels, and sentences without labels were removed from the manual-mm-split.xlsx dataset. The FOMC.xlsx dataset had all duplicate sentences and sentences with an indeterminate growth sentiment label removed. After performing these deduplication and cleansing steps, the manual-mm-split.xlsx and the FOMC.xlsx datasets consisted of 966 and 908 sentences respectively. The manual-mm-split.xlsx, FOMC.xlsx, and S&P 500 datasets all featured imbalanced classes. As can be seen in Table I, the manual-mm-split.xlsx dataset's stance dimension favors the neutral class. The FOMC.xlsx dataset's growth sentiment dimension also favors the neutral class, and the S&P 500 dataset is biased in the upward direction. All of these class imbalances are explainable, as FOMC sentences are typically nuanced with conflicting policy signal. The upward market direction bias aligns with the long-term performance of equity markets from 2006 to 2024.

For RQ1, the manual-mm-split.xlsx dataset was broken up into a 60/20/20 train (579 samples), validation (193 samples), and test (194 samples) split. The enhanced FOMC.xlsx dataset consisting of the merged S&P 500 and macroeconomic features was stratified into a 80/20 chronological split based on FOMC meeting event dates. This temporal split was implemented to prevent time-series-related issues with random shuffling such as look-ahead bias. Furthermore, the enhanced FOMC.xlsx dataset was aggregated at the FOMC meeting date grain to compute an event-level hawkish/dovish stance measure by date. This stance feature was computed to capture the the overall monetary policy signal by FOMC meeting date, as a single meeting date consisted of many sentences with varying tones. Similar GPT-4-labeled growth stance and generic sentiment measures (per-sentence labels) originating from [2] were aggregated at the FOMC meeting level and added for evaluation in the RQ2 pipeline. This aggregation significantly reduced the size of the RQ2 dataset to 104 training samples and 27 test samples.

B. Research Question 1: Stance Classification

Multiple baseline experiments for RQ1 were implemented as control groups to measure the effectiveness of the BERT transformer models and the LLMs in the hawkish/dovish stance classification task. A 1,000 feature term frequency and inverse document frequency (TD-IDF) vocabulary was created and paired with logistic regression (LR) to act as a baseline for

TABLE I
DATA DISTRIBUTION DETAILS

RQ Feature	Class Percentages		
	Class 1	***Class 2***	***Class 3***
RQ1 Stance	Dovish (30.6%)	Hawkish (27.1%)	Neutral (42.2%)
RQ2 Growth	Positive (25.2%)	Negative (24.9%)	Neutral (49.9%)
RQ2 Mkt Direction	Up (61.3%)	Down (38.7%)	N/A

the stance classification task. The baseline TD-IDF LR model obtained an accuracy score of 0.5567 and a macro F1 score of 0.5233. An additional baseline was computed using the same TD-IDF vocabulary and a linear support vector machine (SVM) model. The baseline TD-IDF SVM approach obtained an accuracy of 0.5103 and a macro F1 score of 0.5011. The TD-IDF LR model produced higher accuracy and macro F1 metrics, while the TD-IDF SVM model exhibited marginally more balanced results between the different classes.

The next step in the RQ1 pipeline consisted of training and evaluating multiple BERT transformer models on the stance classification task. Four BERT models were evaluated and compared: distilBERT-base-uncased, BERT-base-uncased, ProsusAI/finBERT, and ZiweiChen/FinBERT-FOMC. The first three BERT models were all trained using a grid search approach over 24 runs to identify the best hyperparameters. The FinBERT-FOMC model was evaluated using a zero-shot approach to evaluate its classification performance with fine-tuned, domain specific weights.

Lastly, five LLMs were evaluated on the stance classification task using a few-shot prompt containing example sentences from each of the hawkish/dovish/neutral classes in the manual-mm-split.xlsx dataset. The following LLMs were evaluated: Claude Sonnet 4.6 (Amazon Bedrock), Claude Opus 4.6 (Amazon Bedrock), Llama 3.1 8B (Groq), Llama 3.3 70B (Amazon Bedrock), and GPT-OSS-120B (Amazon Bedrock). Each LLM labeled the full manual-mm-split.xlsx dataset and also labeled the FOMC.xlsx dataset to add a stance feature for the market direction prediction task for RQ2. The baseline, BERT-family, and LLM classification results were evaluated, and the model with the best metrics was chosen to produce a hawkish/dovish stance feature for the RQ2 (FOMC.xlsx) dataset.

C. *Research Question 2: Market Direction Prediction Pipeline*

Two baseline models were created for the RQ2 market direction prediction task. A simple majority class approach was implemented to compute a result where the upward class was always chosen. The majority class approach produced an accuracy of 0.6667, which is reasonable when considering the overall upward trend of the S&P 500 from the 2006 to 2024 period. An additional baseline valence aware dictionary and sEntiment reasoner (VADER) model was implemented at the sentence level, and the model produced a directional classification accuracy of 0.5179. The baseline majority class approach was used for downstream RQ2 comparisons because it produced a higher accuracy metric than the VADER model.

Expanding beyond the baseline approaches, the FOMC.xlsx dataset was aggregated by FOMC meeting date and enhanced to include a labeled hawkish/dovish stance feature generated by the best-performing classifier from RQ1 and a GPT-4-labeled growth sentiment score feature produced by [2]. The dataset also included inflation and unemployment macroeconomic features obtained from the FRED. The FOMC meeting-level dataset was then used to predict 3-day market direction with multiple feature-set combinations. All the feature-set configurations that were evaluated as part of RQ2 are listed in table II. Logistic regression and random forest models are used for the market direction prediction binary classification task. The classifiers are trained on 104 samples and evaluated on 27 test samples.

TABLE II
RQ2 CONFIGURATIONS

Feature Set
Macro Only
Macro + Generic Sentiment (Sent)
Macro + Growth Stance
Macro + LLM Stance
Macro + Sent + LLM Stance
Macro + Sent + Growth Stance
Macro + Sent + All Stance
Sent + Growth Stance

D. *Fine-Tuning Approach and Hyperparameters*

A grid search approach was utilized to determine the best hyperparameters for each of the BERT models (excluding FinBERT-FOMC) trained on the RQ1 stance classification task. The grid search consisted of four learning rates (3e-5, 2e-5, 1e-5, 1e-4), two batch sizes (16, 8), and three seeds (5768, 78516, 944601), which was modeled after the grid search approach followed by the researchers in [3]. Each BERT run had a maximum of five epochs, with early stopping implemented based on the validation loss. A patience of two was also used to allow the BERT models to continue training in the case of initialization inconsistencies. The maximum sequence length was set to 256 tokens to match the configuration implemented by the researchers in [3].

Regularization techniques were implemented to help prevent the BERT models from quickly overfitting, as the RQ1 train dataset consisted of only 579 sentences. Weight decay was increased to 0.05 from the default of 0.01 to enforce a more significant penalty on larger weight values. The dropout probability remained at the default 0.01 probability, as increased dropout probability significantly reduced convergence speed during test runs. This reduction in convergence speed was unacceptable, as all BERT models were trained locally on a laptop CPU. A class-balanced cross-entropy loss function was used to apply more significant penalties to minority classes (hawkish and dovish) that were misclassified. The optimal

BERT hyperparameters were chosen based on the average validation macro-F1 scores among the three seeds.

The LLM labeling task utilized few-shot prompting with seven example sentences (2 dovish, 2 hawkish, and 3 neutral) and their corresponding stance labels provided in the prompt for reference. The LLM was instructed to classify each FOMC sentence in the RQ1 dataset based on their monetary policy signal, and the prompt provided economic context behind each of the three stance labels. The LLM was instructed to provide one label per sentence and to return the labels as a JSON integer array. Batching was utilized to reduce the total number of API calls, and the batch size was set to ten sentences. The temperature parameter was set to 0.0 to reduce variability in the LLM labeling output. After labeling the RQ1 dataset and evaluating performance, the LLM was instructed to label the full RQ2 dataset to use for the market direction prediction step (if chosen as the classifier for the RQ2 pipeline).

E. Evaluation Metrics

The evaluation metric used for RQ1 was chosen to account for the imbalance present in the dataset that favored the neutral class. For the RQ1 stance classification task, all models were evaluated primarily on their macro F1 scores to provide equal weighting to all three classes in the dataset. The BERT models (excluding FinBERT-FOMC) were evaluated based on their average validation macro F1 scores during the fine-tuning process. The RQ2 market direction classification task was evaluated based on the accuracy score, as correctly predicting the 3-day future market direction was the key task. Additionally, the 10-day future market prediction robustness test was evaluated based on the accuracy metric. The input truncation robustness test was also evaluated on the precision metric.

F. Summary Diagram

A summary process diagram (figure 1) is presented below that details the comprehensive research process and both RQ pipelines.

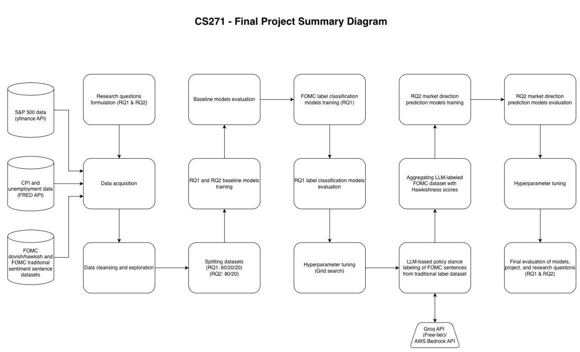

Fig. 1. Research process and experiment flowchart.

III. EXPERIMENTS

As part of the research project, a variety of different experiments were performed in the RQ1 and RQ2 pipelines. Multiple robustness tests were performed to analyze model performance over different data distributions and failure scenarios. Inference efficiency metrics are also included to evaluate the runtime performance of the various models implemented.

A. Experimental Setup

All experiments completed as part of the research project were performed locally on an Apple MacBook Pro M1 with 16GB of RAM, excluding LLM inference. No dedicated GPU was used for this project. All Python code was executed in a local virtual environment using Jupyter Notebook. Python 3.9.x was used on the ARM64 version of macOS.

Several Python libraries were used to perform data acquisition, cleaning, analysis, experiments, logging, and evaluation. The Groq library was used for its free access to a select number of LLMs for the stance classification generation task. The Boto3 library was used for its Converse API to call Amazon Bedrock models for additional LLM inference, as the Groq free tier was found to be insufficient for the LLM labeling required for RQ1. The Transformers library was used to reference BERT transformer models for stance classification and evaluation. MLflow was used to log experiment results such as the RQ1 and RQ2 baseline models, BERT fine tuning, and LLM labeling. Table III displays the most significant software libraries used in the project.

The following experiments were performed as part of the study:

- RQ1 baseline: TF-IDF + LR (stance classification)
- RQ1 baseline: TF-IDF + SVM (stance classification)
- RQ2 baseline: Majority class (3-day market direction prediction)
- RQ2 baseline: VADER compound sentiment (3-day market direction prediction)
- RQ1: distilBERT-base-uncased, BERT-base-uncased, and ProsusAI/finBERT (fine-tuned stance classification)
- RQ1: ZiweiChen/FinBERT-FOMC (zero-shot stance classification)
- RQ1: Claude Sonnet 4.6 (Amazon Bedrock), Claude Opus 4.6 (Amazon Bedrock), Llama 3.1 8B (Groq), Llama 3.3 70B (Amazon Bedrock), and GPT-OSS-120B (Amazon Bedrock) (few-shot prompt stance classification)
- RQ2: LR with various feature-sets (3-day market direction prediction)
- RQ2: Random forests with various feature-sets (3-day market direction prediction)
- Robustness test 1: LR with various feature-sets (10-day market direction prediction)
- Robustness test 2: LR with various feature-sets (3-day market direction prediction by sub-period)
- Robustness test 3: LR (stance classification with input truncation)

IV. RESULTS

The following section evaluates and compares the results obtained for the RQ1 and RQ2 pipelines and the three ro-

TABLE III
SOFTWARE LIBRARIES USED

Library	Purpose
Pandas	Data manipulation
NumPy	Data manipulation
Scikit-learn	Baseline modeling
yfinance	API used for S&P 500 data acquisition
Groq	API used for Groq LLM inference
Boto3	API used for Amazon Bedrock LLM inference
MLflow	Experiment logging
Transformers	BERT family transformer models

bustness tests performed. The section also provides an error analysis and evaluates the inference efficiency of the different models utilized.

A. Research Question 1: Stance Classification

The RQ1 stance classification task delivered some unexpected results, particularly with regard to the performance of the domain-specific ProsusAI/finBERT and ZiweiChen/FinBERT-FOMC transformer models. As can be seen in table IV, both the ProsusAI/finBERT and ZiweiChen/FinBERT-FOMC models performed worse than the TF-IDF + LR baseline, BERT-base-uncased, and distilBERT-base-uncased models in both mean accuracy and macro F1 metrics. Interestingly, Claude Sonnet 4.6 outperformed Claude Opus 4.6 and achieved the best stance classification performance out of all the models tested as part of RQ1. Thus, Claude Sonnet 4.6 was the model used to generate stance labels for the RQ2 dataset.

In [3], the researchers used ChatGPT-3.5-Turbo to perform the same stance classification task using a zero-shot prompt. They obtained an F1 score of 0.5997 with the ChatGPT-3.5-Turbo model. By using a more state-of-the-art LLM (Claude Sonnet 4.6) and few-shot prompting, this study was able to improve LLM stance classification performance by roughly five percent (based on F1 scores). When comparing BERT-family performance, [3] achieved F1 scores of 0.6115 and 0.6486 for BERT-base-uncased and FinBERT respectively. The BERT-base-uncased F1 scores between the two studies are very comparable (0.6115 in [3] and 0.6186), but the FinBERT F1 scores vary widely (0.6486 in [3] and 0.5206). Possible reasons for the poor classification performance could include inadequate fine-tuning for the FinBERT model and the FinBERT-FOMC model being pre-trained on FOMC sentences for traditional positive/negative sentiment classification. This significant classification variance is discussed in greater detail in the limitations section.

The best BERT-family hyperparameter configurations for each model were determined based on the average validation macro F1 scores across the three seeds. The best hyperparameter configurations are displayed in the following list for the three fine-tuned BERT models.

- BERT-base-uncased: Learning rate = 3e-5; batch size = 8

TABLE IV
RQ1 STANCE CLASSIFICATION RESULTS

Model	Accuracy	Macro F1
Claude Sonnet 4.6	0.6598	0.6503
Claude Opus 4.6	0.6546	0.6455
Llama 3.3 70B	0.6340	0.6310
BERT-base-uncased	0.6186	0.6223
GPT-OSS-120B	0.6134	0.6126
distilBERT-base-uncased	0.5670	0.5415
TF-IDF + LR	0.5567	0.5233
ProsusAI/finBERT	0.5206	0.5052
TF-IDF + SVM	0.5103	0.5011
FinBERT-FOMC	0.4588	0.4506

- distilBERT-base-uncased: Learning rate = 1e-4; batch size = 8
- ProsusAI/finBERT: Learning rate = 3e-5; batch size = 16

In summary, LLMs with few-shot prompts were able to outperform small, medium-sized, and domain-specific BERT-family transformer models. Few-shot prompting seems to improve the stance classification performance of LLMs compared to zero-shot prompting. Additionally, domain-specific BERT-family models that are trained to classify FOMC-specific text using traditional positive/negative sentiment labels may not be as well-suited as generalized BERT models for a specialized monetary policy stance classification task.

B. Research Question 2: Market Direction Prediction

The results generated from the RQ2 market direction pipeline are interesting and should be evaluated with caution due to the lack of FOMC meeting-level samples (104 train samples and 27 test samples). This constraint was caused by the researcher's annotation strategies in [2] and [3]. Being that they each only had a few professionals to annotate sentences, it was not feasible to create a fully-annotated corpus of more than 30,000 sentences from FOMC meeting minutes transcripts. This dataset sizing limitation is discussed in further detail in the limitations and future work sections. Bootstrap confidence intervals were computed to analyze the confidence of each model's prediction. As can be seen in table V, all the feature-set configurations had wide-ranging confidence intervals. None of the tested configurations were able to outperform the majority class baseline. All the configurations obtained an accuracy metric between 0.5926 to 0.6667 with overlapping confidence intervals.

In conclusion, the small FOMC meeting-level dataset presented significant limitations when interpreting the RQ2 pipeline results. No feature-set evaluated as part of RQ2 was able to surpass the majority class baseline, and the confidence intervals for each configuration exhibit significant overlap. Thus, the LLM-generated hawkish/dovish stance feature does not present any clear improvement in S&P 500 3-day market direction prediction accuracy.

C. Robustness Testing

In addition to the experiments performed as part of the RQ1 and RQ2 pipelines, three robustness tests were performed

TABLE V
RQ2 3-Day Market Prediction Results

Feature Set	Accuracy	95% CI
Majority Class Baseline	0.6667	[0.481, 0.815]
Macro + Generic Sentiment (Sent)	0.6296	[0.296, 0.667]
LR (Macro + Sent + Growth Stance)	0.6296	[0.296, 0.667]
Sent + Growth Stance	0.6296	[0.481, 0.815]
RF (Macro + Sent + LLM Stance)	0.6296	[0.444, 0.815]
Macro Only	0.5926	[0.296, 0.667]
Macro + Growth Stance	0.5926	[0.296, 0.667]
Macro + LLM Stance	0.5926	[0.296, 0.667]
LR (Macro + Sent + LLM Stance)	0.5926	[0.296, 0.667]
Macro + Sent + All Stance	0.5926	[0.296, 0.667]
RF (Macro + Sent + Growth Stance)	0.5926	[0.407, 0.778]

to analyze results obtained from different subsets of data, prediction windows, and model failure scenarios.

The first robustness test (A) extended the market direction prediction window from three days to ten days. As can be observed in table VI, all tested feature sets saw an improvement in prediction accuracy when increasing the window to ten days. The model consisting of only macroeconomic indicators saw the largest improvement in precision, with an 18.5% delta over its 3-day performance. The model containing the LLM stance feature experienced only a modest improvement of 3.7% in accuracy. These accuracy improvements are explainable, as a wider time horizon reduces the significance of near-term market volatility.

TABLE VI
Robustness Test A Results

3-Day vs 10-Day Prediction Window (LR)		
Model	***3-Day Acc.***	***10-Day Acc.***
Majority Class	0.6667	0.7037
Macro Only	0.5926	0.7778
Macro + Sent + Growth Stance	0.6296	0.7037
Macro + Sent + LLM Stance	0.5926	0.6296

An additional robustness test (B) was performed to analyze the predictive accuracy of two RQ2 feature sets on different temporal subsets of data. Interestingly, the model with the LLM-generated hawkish/dovish stance feature produced equivalent or better accuracy metrics than the majority class in the 2006-2015 and 2016-2019 subsets (table VII). The particular model was less accurate during the COVID 2020-2022 subset, which could be attributed to the unique exogenous shock that the virus introduced into the global economy.

Lastly, a third robustness test (C) was performed to evaluate the stance classification performance of the TF-IDF+LR model used in the RQ1 pipeline on truncated sentences. This test simulated the classifier's performance when being fed corrupted sentences as input. The baseline TD-IDF+LR model was chosen due to its fast inference speed. As shown in table VIII, the classification accuracy of the TF-IDF+LR model declined as the maximum number of characters in input sentences declined.

TABLE VII
Robustness Test B Results

Macro + Sent + Growth Stance (LR)		
Period	***Accuracy***	***Majority Class Acc.***
Pre-2016	0.5890	0.5890
2016-2019	0.5938	0.6250
COVID 2020-2022	0.6250	0.6667
Test Set	0.6296	0.6667
Macro + Sent + LLM Hawkish/Dovish Stance (LR)		
Period	***Accuracy***	***Majority Class Acc.***
Pre-2016	0.6301	0.5890
2016-2019	0.6250	0.6250
COVID 2020-2022	0.5833	0.6667
Test Set	0.5926	0.6667

TABLE VIII
Robustness Test C Results

TF-IDF+LR on Truncated Sentences		
Sentence Length	***Accuracy***	***Delta***
Full Sentences	0.5567	
First 200 Chars	0.5412	-0.0155
First 100 Chars	0.5000	-0.0567
First 50 Chars	0.4278	-0.1289
First 25 Chars	0.4227	-0.1340

D. Error Analysis

As previously mentioned, the FinBERT and FinBERT-FOMC models produced relatively poor stance classification results when compared to the results obtained by [3]. This is most likely due to domain-specific, pretrained weights being trained for other classification tasks. With regard to FinBERT, the model is pretrained on generalized financial text rather than nuanced FOMC sentences. As part of the fine-tuning process for the FinBERT model in this study, its classification head was re-initialized to properly classify the FOMC sentences into one of the three hawkish/dovish/neutral classes. Thus, this classification head re-initialization, combined with five epochs and early stopping may have hindered the FinBERT model's convergence. Five epochs was chosen due to hardware limitations. Similarly, the FinBERT-FOMC model was pretrained on FOMC sentences to perform traditional sentiment analysis. The model's classification head was not re-initialized. Instead, its positive/negative classifications were translated into hawkish/dovish equivalents. This label translation was not entirely accurate, as a sentence could have possessed a traditionally positive sentiment while providing an underlying hawkish signal. Therefore, the label translation approach and the zero-shot training most likely contributed to the FinBERT-FOMC model's poor stance classification performance.

An error analysis was performed on the FinBERT model's stance classification output. Of 194 test samples, the most common error was incorrectly labeling true neutral sentences as dovish (26 instances). There were 22 instances of which the FinBERT model incorrectly classified true hawkish sentences as dovish, and 16 examples of the opposite scenario (true dovish labeled as hawkish). These confusion metrics provided no clear pattern and help to illustrate the difficulty of properly classifying complex FOMC sentences.

E. Inference Efficiency

Several different models were analyzed and evaluated as part of this research project. The main drivers of inference latency were the BERT-family transformer models and the LLMs when performing the RQ1 stance classification task. The distilBERT-base-uncased, BERT-base-uncased, and FinBERT models took 81 minutes to train locally, with each model executing 24 runs as part of the grid search process. The LLMs each took 18-21 minutes to label all sentences in both the RQ1 and RQ2 datasets when calling the Groq or Amazon Bedrock Converse APIs.

The TF-IDF+LR model averaged 2.61 milliseconds (ms) to complete the RQ1 stance classification task, while the TF-IDF+SVM model averaged 16.19 ms. In comparison, the FinBERT-FOMC model, using a zero-shot training approach, averaged about 3 seconds to complete the same task. The LR models used for the RQ2 market direction prediction task averaged 0.023 ms.

V. Conclusion

In summary, this study yielded several interesting insights and observations. A recap of key findings, limitations, and future work are included in this section.

A. Summary of Findings

Based on the RQ1 stance classification results, domain-specific, fine-tuned transformer models were outperformed by the baseline TD-IDF+LR model, generalized BERT models, and LLMs. The FinBERT-FOMC model (zero-shot) exhibited the worst classification performance of all models tested. In contrast, Claude Sonnet 4.6 provided the best stance classification performance, even outperforming the larger Claude Opus 4.6 model. This research project was able to produce improved LLM stance classification metrics compared to [3]. These improvements can most likely be attributed to model improvements and the use of few-shot prompting with example sentences compared to zero-shot prompting used by [3].

In analyzing the RQ2 3-day market direction prediction accuracy results, models with the LLM-generated stance feature did not provide any enhancement in precision. The baseline majority class model produced the highest accuracy of all feature sets tested as part of RQ2. The event-level FOMC dataset used in the RQ2 pipeline was very small, consisting of only 104 train samples and 27 test samples. Thus, it was difficult to produce any definitive takeaways from the market direction prediction results. This uncertainty is further illustrated by the wide confidence intervals produced by each of the models in the RQ2 pipeline.

The three robustness tests performed help to provide additional information. Market prediction accuracy tended to improve when increasing the prediction window from three to ten days. A RQ2 model featuring the LLM-generated stance label seemed to outperform the majority class during the 2006 to 2015 period. Lastly, stance classification performance tended to decrease as input sentences were increasingly truncated.

B. Limitations

This research project consisted of several clear limitations and/or constraints. The dominating constraint was the sizing of both datasets used in the RQ1 and RQ2 pipelines. The dataset used for the RQ1 stance classification task consisted of only 966 samples in total. The RQ2 FOMC event-level dataset featured only 134 total samples, which produced unreliable predictive results with wide-ranging confidence intervals. Additionally, the LLM pseudo-labeling approach utilized in the RQ1 pipeline introduced further instability in the predictive accuracy results for RQ2. The leading model (Claude Sonnet 4.6) achieved an accuracy of 0.6598, which signaled that the LLM-generated stance feature itself was not reliably accurate.

In addition to dataset sizing, local compute resources presented another constraint. The MacBook Pro M1 used to train all the BERT-family models locally could only successfully process a limited number of training runs before experiencing memory exhaustion failure. Thus, the maximum number of epochs per individual run was set to five with early stopping enabled based on validation loss. These hyperparameters helped to control the number of BERT runs, but could have potentially impacted the convergence of the FinBERT model. Additionally, [3] obtained an F1 score of 0.7128 using the RoBERTa-large model for the hawkish/dovish stance classification task. Due to hardware limitations, this study did not attempt to experiment with larger BERT-family models.

Lastly, the Groq free-tier for LLM inference was found to be a limitation. Even with batch prompting implemented to reduce API calls and sleep durations between batches, rate limits were consistently hit before the larger LLMs could complete their RQ1 stance classification tasks. Thus, Amazon Bedrock was used to provide access to serverless LLM inference at a larger scale. After pivoting to Amazon Bedrock, rate limits were much less common and all LLMs were able to complete their required classification tasks.

C. Future Work

This study highlights key opportunities for future work and experimentation. To address the lack of human-annotated FOMC sentences, LLMs can be used for data augmentation purposes to generate synthetic sentences and for pseudo-labeling. The researchers in both [2] and [3] had to reduce their dataset sizes dramatically due to human annotation constraints. An LLM could be used for pseudo-labeling to create a larger corpus to potentially enhance model training. Furthermore, prompting an LLM to create synthetic data (FOMC sentences) could prove useful by producing additional samples for training and helping to balance dataset classes.

Another area of future work could be in LLM prompt optimization, as this research project had success with few-shot prompting as part of the RQ1 stance classification task. Opportunities for improvement exist in identifying few-shot example sentences that are more representative of each class and token optimization to reduce potential rate-limiting. Lastly, further evaluation of robustness test B could be useful in understanding why the model containing the LLM-generated stance feature outperformed the majority class from 2006 to 2015 in 3-day market prediction accuracy.

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